Stopping time and force

antocciphysics | TEACHER KEY | CPA

Learning target: Explain how a longer stop can reduce average force.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 1; 3 course days later.*
Change magnitude 15 kg·m/s; force 7.5 N opposite initial motion.

02. Explain a worked example (worked)
First is four times taller and one quarter as wide; stopping force opposes motion.

03. Look and explain (model)
The wider, lower pulse; equal area means equal impulse.

04. Build the solution (practice)
20 N and 5 N.

05. Use the idea (reason)
External interaction supplies impulse. Cushion extends stopping time/distance; total momentum change for a given stop remains the same.

06. Choose a method (mixed) *Review: Unit 2, Day 8; 12 course days later.*
10 m/s² downward. Timing resolution, sensor fit or release timing can contribute uncertainty.

07. Exit check (exit)
Half as large.

08. STRETCH: Apply it another way (stretch)
Impulse same, average force halves. Peak depends on force-time shape and is not fixed by average alone.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 1; 3 course days later.*
$J=-15$ N·s; $F_{avg}\approx-7.5$ N, left.

02. Explain a worked example (worked)
First is four times taller and one quarter as wide; stopping force opposes motion.

03. Represent and interpret (model)
The wider, lower pulse; equal area means equal impulse.

04. Solve and justify (practice)
Impulse magnitude 1.20 N·s; forces 120 N and 24 N, opposite initial motion.

05. Analyze the reasoning (reason)
External interaction supplies impulse. Cushion extends stopping time/distance; total momentum change for a given stop remains the same.

06. Choose a method (mixed) *Review: Unit 2, Day 8; 12 course days later.*
$g=\Delta v/\Delta t=2.94/0.30=9.8$ m/s². Several points reveal a trend and reduce sensitivity to one noisy observation.

07. Exit check (exit)
$J=-8$ N·s; $F=-16$ N; new $F=-8$ N.

08. STRETCH: Apply it another way (stretch)
Impulse -3 N·s; average -100 N and -25 N. A negative triangle of base 0.12 s needs peak magnitude 50 N.